Weighted Integer Lipschitz Heights: Rare-Event Transfer and Dirichlet Green Certificates

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Abstract

We examine the claimed Dirichlet Gaussian free-field limit of triangular-lattice integer Lipschitz heights with fixed edge weight $x \in [1/\sqrt{2}, 1]$. Exact face-area weights, the full stopped spin/status record, rare-pin normalization, and the boundary condition are retained. The source already states exponential tails and proves explicit constants by midpoint amplification. Using those tails in its subsequent second-moment transfers sharpens the gap-closing error from a square-root loss to $p \log(1/p)$, and the collision-coupling loss to a linear failure probability times an explicit logarithmic polynomial. Separately, we prove a conditional Dirichlet certificate separating bulk residuals, logarithmic contact errors, boundary errors, and unmeasured energy components. It gives finite Wick and characteristic-function error bounds, without equating covariance matching with Gaussianity. These are source-specific local consequences and conditional extensions, not a new global proof, an effective mesh rate, or a uniform theorem in x. Contribution assessment: C3/5, sharper rare-event transfer and a boundary-aware finite Wick certificate give a notable local increment while the global probabilistic inputs remain inherited.

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1 Model, normalization, and scope

Let $\mathbb{T} = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. The face centers of the hexagonal lattice are $\mathbb{T}$. Fix a bounded simply connected C^2 Jordan domain D and inside approximations D_δ bounded by simple hexagonal cycles. Every compact subset is eventually inside; boundary parametrizations converge uniformly to a Jordan parametrization of ∂D. This is more than Hausdorff convergence. It does not bound discrete boundary length or the number of repeated visits of its inner face walk.

For one fixed $x \in [1/\sqrt{2}, 1]$, the source law is

$$\mathbb{P}_{\delta,x}(h) = Z_{\delta,x}^{-1} x^{N(h)}, \qquad h : \mathcal{F}_\delta \to \mathbb{Z}, \quad h|_{\partial \mathcal{F}_\delta} = 0, \quad |h(u) - h(v)| \le 1 \quad (u \sim v). \tag{1}$$

Here $N(h)$ counts each unordered edge with unequal endpoints once. Extend h constantly on each hexagonal face and by zero outside the approximation. The physical pairing is exactly

$$h_\delta(f) = \sum_u w_{\delta,f}(u) h(u), \qquad w_{\delta,f}(u) = \int_{Q_{\delta,u}} f(z)\, \mathrm{d}z. \tag{2}$$

For a test supported inside the approximation,

$$\sum_u w_{\delta,f}(u) = \int f, \qquad \sum_u |w_{\delta,f}(u)| \le \|f\|_1, \qquad |w_{\delta,f}(u)| \le \frac{\sqrt{3}}{2} \delta^2 \|f\|_\infty .$$

Thus zero integral removes the additive lift exactly, at every mesh. Replacing these weights by point samples need not preserve this cancellation.

Write $G_D = (-\Delta_D)^{-1}$ for the negative Euclidean Laplacian with zero Dirichlet trace. Its singularity is $(2\pi)^{-1} \log(1/|z-w|)$. In the Fourier convention $\widehat{f}(p) = \int e^{-ipz} f(z)\, \mathrm{d}z$, the source's plane covariance is

$$\frac{v(x)}{(2\pi)^2} \int_{\mathbb{R}^2} \frac{\widehat{f}(p)\overline{\widehat{g}(p)}}{|p|^2}\, \mathrm{d}p, \qquad \int f = \int g = 0, \tag{3}$$

with $v = (2\pi)^2 c > 0$ for its spectral density coefficient c. The claimed limiting law is $\sqrt{v}\, \Phi_D$, not a counting-measure Green field and not a logarithmically rescaled height average. No explicit $v(x)$ or uniform positivity in x is obtained here.

Throughout, "source" means the pinned September 25 manuscript [1]. Its theorem is attributed, not independently certified by reading all its text. Our local results below either extract stronger bounds from precisely identified source premises or explicitly assume a residual certificate. The pre-existing qualitative GFF assertion is not counted as a contribution.

2 Proof dependencies that must survive reformulation

The following map covers all eight scientific sections; exact line ranges and hashes are in the separate read ledger.

Source section	Indispensable content and boundary of our use
02: discrete	Gray lift, complete-history association, open-cell cuts, online kernels, multiplicative rare-pin comparison, protected offset switches. Crossing input is imported.
03: moments	Shell tails, common-loop covariance, weighted rounded midpoints, central mass in each admissible class, exact area interpolation. We extract its exponential-tail conclusion.
04: plane	Site reflection positivity, positive normal transfer, Cauchy mixture, dispersion defect, endcaps and scalar cutoff fixing v. We retain these as source inputs.
05: pins	Integer bracket traces, marker checks, parity-class moments, rare-probability-normalized reflection norms, five analytic moves, phase projection and gap closing.
06: boundary	Synthetic traces, genuine boundary attachments, stopped arches, absolute-height spectators and vanishing references. Geometry alone does not supply their laws.
07: limit	Tightness and uniform integrability; harmonic moments; removal of diagonal distributions; conditionally independent plane matching; collision coefficients; Wick recursion and law uniqueness.
08: spectra	Scalar spectral measures of commuting unitaries and positive contractions, including endpoint atoms and the positive-exponent analytic convention.

2.1 The conditioning is part of the theorem

The height residues $0, 1, 2, 3$ have Gray states $(B, W) = (+,+), (+,-), (-,-), (-,+)$. On oriented edges,

$$H_v - H_u = \frac{W_u B_v - B_u W_v}{2}.$$

For a full triangular interaction cell, $q = x^2$; for a clipped two-site cell, $q = x$. An open cell has constant first spin and weight q, leaving the complementary spin unrestricted. A closed homogeneous cell has weight $1 - q$; a closed first-spin-nonconstant cell requires constant complementary spin and has weight q. On a prescribed open-cell circuit, crossing factors are constant. Exact cutting also requires that no imposed complementary equality edge crosses the circuit.

Every observed status is accompanied by all its first-spin sites. Retained complementary labels stay unobserved. Merely observing a closed triangle does not preserve the association used in the proof: its first-spin pair covariance is $(1-2q)/(1+2q) < 0$ for $q > 1/2$. Likewise, a geometric circuit without its prescribed status is not the same conditional law. The source's online coupling samples recursively from each coordinate's own-history kernels; it is not an arbitrary pre-sampled ordered coupling conditioned afterwards.

If $\mathcal{R}$ includes the full exterior absolute-height record, the needed plane copy P satisfies

$$\mathbb{P}(P \in A \mid \mathcal{R}) = \nu(A) \quad \text{almost surely, for every } A. \tag{4}$$

This includes unsuccessful searches. It implies independence from all exterior spectators. The unconditional equality $\mathcal{L}(P) = \nu$ is insufficient: if S is a fair sign and $P = S$, then P has the fair-sign marginal but $\mathbb{E}[PS] = 1$, not 0. None of our improved error bounds changes this requirement.

2.2 Limits and singularities

The source fixes x, test supports, pin groups, buffers and windows before $\delta \downarrow 0$. Gap and window closure and the number of boundary references are subsequent limits. Constants may deteriorate when the geometry degenerates. The source proves tightness in H^{-t}_{loc} for $t > 1$ and all fixed-test moments before identifying the limit.

Its no-diagonal argument is stronger than a formal pointwise bound: radial mollification agrees exactly with the separately harmonic function off collision tubes. Tube volume is $O(h^2)$ and its local L^1 contribution is $O(h^2(1+\log(1/h))^{j/2})$. This excludes distributions supported on diagonals. Logarithmic coefficients are then identified using (4); only after their removal can a zero-boundary maximum principle identify the full moment. A finite PSD covariance matrix proves none of these facts.

3 Exponential tails already contained in the source

The source's midpoint-amplification lemma explicitly states both an exponential tail and all finite moments; its proof also gives the constants below. We retain that existing statement and specify where its constants remain uniform. Our increment is the sharper use of this already available tail in the source's later second-moment transfers, not rediscovery of exponential concentration.

Lemma 3.1 (Source tail and permitted uniformity)**.** *Suppose the finite weighted height array admits at most J midpoint classes, with unconditional central mass at least $c > 0$ in $|Y| \le M$ in every class. Suppose rounded outputs are admissible with nondecreasing product weight and $|Y(m_\pm) - [Y(s)+Y(t)]/2| \le M$. Then*

$$\mathbb{P}(|Y| > t) \le 4e^{-t/B}, \qquad B = \frac{2^{L+1}}{\log 2} M, \quad L = 3 + \lceil 4J/c \rceil. \tag{5}$$

The estimate survives limits in distribution. Under the source's stated cut and midpoint premises, zero-total plane tests have fixed B; interior radius-r true-field tests have $B \le C(1+\log(1/r))^{1/2}$. Common-offset gap laws and filled brackets have a uniform B at fixed test geometry, for all sufficiently small gaps after the mesh limit. Finite sums or differences retain exponential tails, with changed constants.

Proof. For $j \ge 3$ put $I_j = [(2^j-4)M, (2^{j+1}+4)M]$ and $p_j = \mathbb{P}(Y \in I_j)$. Some class carries at least p_j/J. Its tail arrays and central arrays have both midpoints in I_{j-1}, so the discrete midpoint inequality gives $p_j \le (J/c)p_{j-1}^2$ for $j \ge 4$. The intervals have overlap at most two. Among $4, \dots, L$ one therefore has $(J/c)p_{j_0} \le 1/2$. Squaring successively and summing the dyadic positive and negative tails gives $4\exp[-(\log 2)2^{-L-1}t/M]$. This is exactly the source's argument, not a new concentration theorem.

The source's zero-trace second moment supplies one class and $M = C(1+\sqrt{\log(n/a)})$. Zero-total plane tests have a fixed central interval after a stopped cut, and their anchored arrays again form one class. General bounded traces follow by height order for nonnegative weights and then splitting signed weights. A union bound for a finite sum transfers exponential tails without assuming independence. For checked separated pins, the protected switches provide mass in every offset-parity class, with $J \le 2^{m-1}$ and constants depending on separation. Crucially, after imposing the common offset, all offsets are zero: the source's gap-closing proof supplies one class and central mass uniform in the closing gap. It is this independent argument, not a uniform limit of separated-pin constants, that gives the last assertion. Distributional limits preserve the tail bound by applying the open-set Portmanteau inequality to $\{|Y| > t\}$. □

Theorem 3.2 (Rare-event product modulus)**.** *Let $Y_1, \dots, Y_k$ be real or complex random variables on any common probability space, with $k \ge 1$ and*

$$\mathbb{P}(|Y_i| > B_i u) \le A_i e^{-u} \quad (u \ge 0), \qquad B_i > 0, \quad A_i \ge 1.$$

No independence is assumed. For any event E of probability at most $p \in (0,1]$, set $A = \sum_i A_i$, $L = \log(A/p)$ and

$$Q_k(L) = k! \sum_{j=0}^{k} \frac{L^j}{j!}. \tag{6}$$

Then

$$\mathbb{E}\left[\mathbf{1}_E \prod_{i=1}^{k} |Y_i|\right] \le \left(\prod_i B_i\right) p Q_k(L). \tag{7}$$

For $k = 1$ this is $B_1 p[1 + \log(A_1/p)]$. The order $p(\log(1/p))^k$ cannot in general be reduced under only these tail assumptions.

Proof. If $Z = \prod_i |Y_i|/B_i$ exceeds u^k, one factor exceeds u. Thus $\mathbb{P}(Z > t) \le A \exp(-t^{1/k})$. Layer cake, with the event retained, gives

$$\mathbb{E}[Z\mathbf{1}_E] \le \int_0^\infty \min\{p, Ae^{-t^{1/k}}\}\,\mathrm{d}t = pL^k + Ak \int_L^\infty u^{k-1} e^{-u}\,\mathrm{d}u.$$

Repeated integration by parts evaluates the last expression as $pQ_k(L)$. For sharp order, take one exponential random variable Z_0 of mean one, set $Y_i = B_i Z_0$, and $E = \{Z_0 > \log(1/p)\}$. The expectation equals $p(\prod_i B_i) Q_k(\log(1/p))$. Perfect dependence is allowed; the union bound's factor k affects constants, not the sharp order. □

3.1 Three source-specific consequences

Use total variation $\|\mu - \nu\|_{\mathrm{TV}} = \sup_A |\mu(A) - \nu(A)|$. A maximal coupling then mismatches with this probability.

Corollary 3.3 (Gap, collision and boundary errors)**.** *Retain all the source's corresponding probabilistic hypotheses.*

1. *In source 05, let p bound the total variation between the common-offset gapped and filled laws on the joint test increment information. If $|F| \le C_F$ and the insertion in both laws has tails $Ae^{-t/B}$ with common A, B, then*

$$|\mathbb{E}_\mu[XF] - \mathbb{E}_\nu[XF]| \le 2C_F B p[1 + \log(A/p)]. \tag{8}$$

   Consequently its mesh-first gap error is $O(\omega(a)\log(e/\omega(a)))$, rather than the $O(\sqrt{\omega(a)})$ obtained by optimizing the source's second-moment truncation.

2. *In the source's j-point collision comparison ($j \ge 2$), suppose the search-plus-mismatch event has mesh-first probability at most Cr^κ. With the same conditionally fixed plane marginal, the difference of the actual and factored plane terms is*

$$O\left(r^\kappa (1 + \log(1/r))^{j+1}\right). \tag{9}$$

   This improves the displayed $O(r^{\kappa/2}(1+\log(1/r)))$ in source 07 for each fixed j, without changing its unknown positive κ.

3. *At N fixed boundary-arch scales, an exterior product of $k-1$ comparable-depth or fixed averages has failure contribution*

$$O\left(e^{-cN}(1+N)^k\right), \tag{10}$$

   in place of the source's second-moment $O(e^{-cN/2})$. For bracket means, $k = 1$.

These bounds have the source's order of limits, not a uniform mesh rate or a computable mesh threshold.

Proof. Couple the joint increment information in the first assertion; both X and F agree off its mismatch event. Apply Theorem 3.2 with $k = 1$ to the two insertions separately. The uniform one-class tail from Lemma 3.1 is valid as the gap closes. This avoids dividing an additive error by a rare first-color pin probability.

For the second assertion, use precisely the source's common coupling of (U, V, F) and (U', V', F). Equation (4) is required to factor $\mathbb{E}[U'V'F]$. Each product has j height factors. The true local U, V have tail scales $O(\sqrt{1 + \log(1/r)})$, all fixed spectators have fixed scales, and the plane zero-total U', V' have fixed scales. Apply (7) separately to both products on the same failure event. Its polynomial is $O((1 + \log(1/r))^j)$ and the first product of scales is $O(1 + \log(1/r))$. This proves (9). Each finite r is fixed before the mesh tends to zero; continuity and monotonicity of the event modulus justify the limsup version.

For the third assertion the source's separate increasing and decreasing searches give a conditional sign of the retained average. The negative contribution is bounded by the product on the failure event. Bounded-trace and comparable-depth smearings have uniformly bounded exponential scales by the one-class midpoint proof and height order. Apply (7) to these k factors, and to positive and negative parts of the entire spectator product as in the source. A finite sum of failure events only changes constants. No moments conditional on an arch failure or random large trace are required. □

Remark 3.4. For instance $Q_2(L) = L^2 + 2L + 2$ and $Q_4(L) = L^4 + 4L^3 + 12L^2 + 24L + 24$. The gain is in the power of a verified failure probability, not necessarily a small practical constant: the midpoint scale in (5) can be enormous. The exponent κ and all geometric constants still come from the source's genuine cuts and crossings. Improving those inputs is a separate problem.

4 A boundary-aware Green defect certificate

The preceding result sharpens an actual error estimate in the source. The next result has a different status: it is a conditional certificate for approximate moment data. The lattice source does not supply numerical bounds for all its inputs. We use the weak inverse and boundary-lifting discipline of the freshly read original private manuscript [7], while crediting the underlying coercive inverse as classical.

Let $V = H_0^1(D; \mathbb{R})$ with $\|u\|_V^2 = a(u, u) = \int_D |\nabla u|^2$. Its dual norm is the energy-dual norm, not the coordinate Euclidean norm. The weak inverse $\mathcal{G}_D : V' \to V$ is defined by $a(\mathcal{G}_D f, w) = \langle f, w\rangle$ for all $w \in V$. It obeys

$$\|\mathcal{G}_D f\|_V = \|f\|_{V'}, \qquad \langle f, \mathcal{G}_D g\rangle = a(\mathcal{G}_D f, \mathcal{G}_D g). \tag{11}$$

For $b \in H^{1/2}(\partial D)$ let H_b be its weak harmonic extension. If b is also bounded, the weak maximum principle gives $\|H_b\|_\infty \le \|b\|_\infty$. All functionals and pairings in the next theorem are real; signed residuals are allowed.

Theorem 4.1 (Separated contact, bulk and boundary errors)**.** *Fix distinct $y_2, \dots, y_j \in D$, real target contact strengths a_i^*, and actual strengths $a_i = a_i^* + d_i$. Suppose the approximate first-variable moment U has the renormalized remainder*

$$w = U - \sum_{i=2}^{j} a_i G_D(\cdot, y_i) \in H^1(D), \qquad \operatorname{Tr} w = b \in H^{1/2}(\partial D) \cap L^\infty(\partial D), \tag{12}$$

and $-\Delta w = R \in V'$ *weakly on* all *of* D*. For every smooth compactly supported* f*,*

$$\int_D fU - \sum_{i=2}^{j} a_i^*(\mathcal{G}_D f)(y_i) = \sum_{i=2}^{j} d_i(\mathcal{G}_D f)(y_i) + \int_D fH_b + \langle R, \mathcal{G}_D f\rangle, \tag{13}$$

$$\left|\int_D fU - \sum_{i=2}^{j} a_i^*(\mathcal{G}_D f)(y_i)\right| \le \sum_{i=2}^{j} |d_i|\,|\mathcal{G}_D f(y_i)| + \|f\|_1\,\|b\|_\infty + \|f\|_{V'}\,\|R\|_{V'}\,. \tag{14}$$

In the source application $a_i^* = vT_{j-2}(y_2, \dots, \widehat{y_i}, \dots, y_j)$*. The hypothesis* (12) *explicitly excludes extra collision-supported distributions and unremoved logarithmic terms. It cannot be inferred from off-diagonal residual samples.*

Proof. Subtract H_b from w. The difference belongs to V and solves $-\Delta(w - H_b) = R$ on D, so coercivity gives $w = H_b + \mathcal{G}_D R$. Integrate against f. Symmetry of the weak inverse gives $\int f\mathcal{G}_D R = \langle R, \mathcal{G}_D f\rangle$, and the Green kernel represents the point-source terms when tested by this smooth f. Cauchy–Schwarz in (11) and the maximum principle give (14). A logarithmic point-source Green function is not itself in $H^1(D)$ in two dimensions; it is precisely the subtraction in (12) that makes the energy argument valid. □

Worked disk certificate. On the unit disk let a single contact error have strength d at zero, let $R = \rho$ be constant and let $b = \beta$ be constant. The entire error is exactly

$$e(z) = \frac{d}{2\pi}\log\frac{1}{|z|} + \beta + \frac{\rho}{4}(1 - |z|^2). \tag{15}$$

For the unit-area average $f = 1/\pi$ on the disk, an L^2 test to which the identity extends by approximation,

$$\int fe = \frac{d}{4\pi} + \beta + \frac{\rho}{8}, \qquad \|f\|_{V'}^2 = \frac{1}{8\pi}, \qquad \|R\|_{V'}^2 = \frac{\pi\rho^2}{8}.$$

The bulk term in (14) is therefore exactly $|\rho|/8$. The three errors add with their signs before taking absolute values. Setting $\rho = 0$ does not remove the contact error or the constant boundary error. This example uses the actual two-dimensional Euclidean normalization, not a one-dimensional inverse as a substitute.

4.1 What a finite energy bank does and does not certify

Choose linearly independent $\psi_1, \dots, \psi_m \in V$ and put $K_{ab} = a(\psi_b, \psi_a)$. For $R \in V'$ write $r_a = \langle R, \psi_a\rangle$. The energy-orthogonal projection of $\mathcal{G}_D R$ onto their span is $\sum_a (K^{-1}r)_a \psi_a$. Pythagoras gives

$$\|R\|_{V'}^2 = r^T K^{-1} r + \|(I - P)\mathcal{G}_D R\|_V^2\,. \tag{16}$$

This is a classical projection identity. It implements the full-residual requirement in the original physical-metric chapter [8]. The measured term is a *lower* bound, not an upper bound for the full residual. One must independently bound the omitted component to use Theorem 4.1.

For smooth sources f, g let $u_f = \mathcal{G}_D f$, $u_{f,m} = Pu_f$ and $e_f = u_f - u_{f,m}$. The finite Green covariance obeys the useful exact identity

$$\langle f, \mathcal{G}_D g\rangle - a(u_{f,m}, u_{g,m}) = a(e_f, e_g), \qquad |a(e_f, e_g)| \le \|e_f\|_V\,\|e_g\|_V\,. \tag{17}$$

Both cross terms vanish because both approximations use the same energy projection. In particular the diagonal finite covariance is lower than the exact one. This remains a deterministic Green

calculation; it says nothing about the law of the height field. If $f = \sum s_a \psi_a$, its load vector is Ms, $M_{ab} = \int_D \psi_a \psi_b$, so the source-to-solution matrix is $K^{-1}M$. The covariance of two coefficient sources is $s^T M K^{-1} M t$, not $s^T K^{-1} t$.

An omitted direction can be arbitrarily large: in any infinite-dimensional V, take $u \neq 0$ perpendicular to the bank and let $R = -\Delta u$. Every r_a is zero but $\|R\|_{V'} = \|u\|_V > 0$. Increasing arithmetic precision in the same bank does not fix this obstruction.

5 Finite Wick errors and an actual law certificate

We now integrate the source's Green recursion against spectators. This is a finite, conditional extension, not a claim that the source already gives effective residual sizes.

Fix real smooth tests $f_1, \dots, f_d$ and let $Y_i = H(f_i)$ have all moments needed below. Define

$$C_{ij} = v\langle f_i, \mathcal{G}_D f_j \rangle.$$

For an ordered list $I = (i_1, \dots, i_k)$ write $M_I = \mathbb{E} \prod_{a=1}^k Y_{i_a}$; repeated indices are allowed and $M_\varnothing = 1$. Assume a one-variable distribution u_{I^-} representing $\mathbb{E}[H(z) \prod_{a=2}^k Y_{i_a}]$ satisfies

$$-\Delta u_{I^-} = v \sum_{a=2}^k f_{i_a} M_{I\setminus\{1,a\}} + R_I, \qquad u_{I^-} = H_{b_I} + \mathcal{G}_D \left(v \sum_{a=2}^k f_{i_a} M_{I\setminus\{1,a\}} + R_I \right). \tag{18}$$

Here $R_I \in V'$, $b_I \in H^{1/2} \cap L^\infty$, and the second identity states the global boundary closure, not just the PDE away from contacts. All collision errors after spectator integration are included in R_I. If obtained from Theorem 4.1, this requires integrability of its bounds in the spectator variables; pointwise estimates at separated tuples alone do not give it. These are assumptions on continuum or suitably regularized moment data. A raw piecewise constant lattice moment need not belong to H^1, and its Laplacian need not lie in V'. A smoothing procedure would need its own error account before this certificate could be applied.

Put

$$\eta_I = \|f_{i_1}\|_{V'} \|R_I\|_{V'} + \|f_{i_1}\|_1 \|b_I\|_\infty. \tag{19}$$

Testing (18) by $\mathcal{G}_D f_{i_1}$ proves

$$\left| M_I - \sum_{a=2}^k C_{i_1 i_a} M_{I\setminus\{1,a\}} \right| \leq \eta_I. \tag{20}$$

For $k = 1$ the sum is empty. Centering is thus an explicit channel, not an automatic consequence of a covariance fit.

Proposition 5.1 (Error propagation without a Gaussian assumption)**.** *Let W_I be the centered Gaussian Wick moment with covariance C. If (20) holds through degree K, errors can be certified recursively by*

$$E_\varnothing = 0, \qquad E_I = \eta_I + \sum_{a=2}^k |C_{i_1 i_a}| E_{I\setminus\{1,a\}}, \qquad |M_I - W_I| \leq E_I. \tag{21}$$

For a single real variable $Y = H(f)$ with target variance $s = v\|f\|_{V'}^2$, scalar defects η_k give

$$E_{2m} = \sum_{q=1}^m s^{m-q} \frac{(2m-1)!!}{(2q-1)!!} \eta_{2q}, \tag{22}$$

$$E_{2m+1} = \sum_{q=0}^m s^{m-q} \frac{(2m)!!}{(2q)!!} \eta_{2q+1}. \tag{23}$$

The formulas include $s = 0$ with the polynomial convention $0^0 = 1$ in the equal-index terms.

Proof. Subtract the exact Gaussian Wick recursion from (20) and use the triangle inequality. Induction yields (21). In one variable it reads $E_k = \eta_k + (k-1)sE_{k-2}$. Iterating gives the displayed products of consecutive odd or even integers. This proof concerns actual moments of a possibly non-Gaussian random variable, not a presupposed Gaussian field. □

Theorem 5.2 (Finite characteristic-function enclosure)**.** *Suppose the preceding scalar error bounds are valid through degree $2m$, $m \geq 1$, for the actual variable Y. Then for every real t,*

$$\left|\mathbb{E}e^{itY} - e^{-st^2/2}\right| \leq \sum_{k=1}^{2m-1} \frac{|t|^k E_k}{k!} + \frac{|t|^{2m}}{(2m)!}\left(2(2m-1)!!s^m + E_{2m}\right). \tag{24}$$

This is a finite-frequency certificate. For a vector, apply it to $Y = \sum_i a_i H(f_i)$ only when the required defects have been verified for that linear combination. A finite list of directions is not an all-directions field law.

Proof. Taylor's formula with integral remainder on the real line gives $|e^{ity} - \sum_{k=0}^{2m-1}(ity)^k/k!| \leq |ty|^{2m}/(2m)!$. Apply it to Y and to a centered Gaussian Z of variance s. The polynomial differences are bounded using E_k. Moreover $\mathbb{E}|Y|^{2m} = M_{2m} \leq (2m-1)!!s^m + E_{2m}$, while $\mathbb{E}|Z|^{2m} = (2m-1)!!s^m$. Summing the two remainder bounds proves (24), including degenerate $s = 0$. No moment determinacy or unverified exponential moment is needed for this finite estimate. □

Covariance matching fails this stronger test. Take $Y = \pm\sqrt{s}$ with equal probabilities and $s > 0$. Its mean and covariance equal those of the target Gaussian, but

$$M_4 - 3sM_2 = -2s^2.$$

Thus any fourth-order defect budget must be at least $2s^2$. A covariance-only finite Gram certificate misses exactly this obstruction. More generally one can add a random harmonic constant to an interior field: zero-total interior tests and off-diagonal harmonicity do not detect the added boundary mode. The boundary term in (19) is not optional.

When convergence follows, and when it does not. If the defects vanish for every fixed order and every fixed test combination, then taking the limit in (24) first at fixed m and subsequently sending $m \to \infty$ identifies all those characteristic functions: the Gaussian remainder is $2(st^2/2)^m/m! \to 0$. Distributional tightness remains a separate requirement for a field limit. This reproduces a conditional route to law identification, not a rate from a finite matrix. The source supplies its own all-order collision and boundary arguments instead of the quantitative residual hypotheses in (18).

6 Comparisons, costs, and future work

What is retained. The source's probability law, endpoint weight, exact area convention and Dirichlet normalization remain unchanged. Its complete-history restrictions, integer offset checks, actual boundary attachment and conditional plane independence are preserved. We found no correction to assert within the locally checked chain; this is not a full independent verification of all global deductions or cited proofs. The improvement is stronger rare-event bookkeeping using a tail already justified by that chain, and a separate conditional defect certificate adapted to its collision recursion.

Original private proof comparison. The fresh read of [7], lines 65–197, includes the coercive weak inverse, boundary lifting, PDE–Green equivalence and source-measure convention. These support (13), not a probabilistic Gaussianity assertion. The complete physical-metric chapter [8], lines 1–210, proves physical projection, omitted residual and finite-measurement limitations. Its included figure and cited earlier reports were not opened; they are not claimed as fresh evidence. Neither private source replaces the source's spin crossings, reflection positivity, offset synchronization or boundary arches. No earlier report's local contribution is counted again here, and no whole-corpus read is claimed.

Public status checked October 9, 2026. Primary abstracts and metadata were freshly checked, not their complete proofs. Glazman–Manolescu [2] establishes logarithmic fluctuations and an infinite-volume loop law for uniform triangular-lattice heights; its v3 revision is February 27, 2023. Glazman–Lammers [3] establishes delocalization and logarithmic roughness in the relevant weighted regime; the cited v3 is October 27, 2023, and the journal publication is 2025. Glazman–Harel–Zelesko [4], v3 April 19, 2026, treats macroscopic loops in a broader $O(n)$ region. These statements are not themselves the weighted Dirichlet GFF theorem. Duminil-Copin–Kozlowski–Lammers–Manolescu [5], March 6, 2026, proves a full-plane six-vertex GFF result in its specified square-lattice regime, not this boundary-weighted triangular model. The cached manuscript's asserted theorem is kept distinct from these external results and from its two-arc companion.

Advantages and costs. The rare-event theorem needs no independence among height factors and preserves rare-pin normalization; it gains the full failure exponent at the expense of logarithmic factors and potentially very large fixed constants. The Green certificate separates physically different failure modes and tolerates signed, non-Gaussian approximate data. Its costs are substantial: a full dual residual, boundary trace and all collision terms must be justified. The measured finite part alone is insufficient; high-order defects can grow combinatorially. Neither result proves a computational speedup, an effective mesh threshold, or uniform control as x or the geometry varies.

Feasible extensions. Useful next steps are quantitative source bounds for the normalized reflection seminorm and a certified omitted-energy bound for selected smooth tests. A numerical Green solve could then contribute to (19), but it would still need an independently controlled probabilistic residual. The product modulus applies to varying tail scales and finitely many mixed-sign spectators. Improving the midpoint constant, or proving better-than-exponential tails in the relevant pinned ensemble, would reduce its practical cost. Such stronger concentration and a joint mesh/window schedule are open tasks here, not conclusions.

7 Verification and evidence boundaries

All fourteen cached public text files were literally displayed in bounded, untruncated chunks, including every scientific section, the bibliography and TikZ figure sources. Their pre-read hashes were captured and persisted before those reads. The original private reads total 343 lines in two files; the remainder of the 1066-line PDE manuscript is explicitly unread. Private material was not sent to a public search service. The read ledger records exact display intervals, unread complements and matching endpoint hashes. Parent captured-tree authentication is provenance only.

The owned standard-library executable checks passed 1403 cases and 4624 assertions using exact rational arithmetic and polynomial coefficients for the product-tail polynomial, weighted midpoint energy and rounding, Gray lifts and closed-cell covariance, physical Green projection defects, disk

normalization, Wick error propagation and the conditional-marginal counterexample. Case counts and assertion counts are reported separately in the test receipt. They are finite algebraic fixtures, not tests of the source's infinite-volume GFF assertion, crossing probabilities or uniform analytic constants. Public external proofs remain imported at the specific scope stated above. The final build, individual-page visual receipt, contribution assessment and freeze manifest are separate owned artifacts.

References

[1] OpenAI. *Gaussian free-field limits of weighted integer Lipschitz heights*. September 25, 2026. Pinned source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; public manuscript source. Citations to source sections refer to this captured text.

[2] A. Glazman and I. Manolescu. *Uniform Lipschitz functions on the triangular lattice have logarithmic variations*. Communications in Mathematical Physics 381 (2021), 1153–1221. arXiv:1810.05592v3.

[3] A. Glazman and P. Lammers. *Delocalisation and continuity in 2D: loop $O(2)$, six-vertex, and random-cluster models*. Communications in Mathematical Physics 406 (2025), Article 108. arXiv:2306.01527v3; source crossing locator: Appendix A, equation (55).

[4] A. Glazman, M. Harel and N. Zelesko. *Planar percolation and the loop $O(n)$ model*. arXiv:2508.20917v3, 2026 revision.

[5] H. Duminil-Copin, K. K. Kozlowski, P. Lammers and I. Manolescu. *Gaussian free field convergence of the six-vertex model with $-1 \leq \Delta \leq -1/2$*. arXiv:2603.06268v1, 2026.

[6] N. Gozlan, C. Roberto, P.-M. Samson and P. Tetali. *Transport proofs of some discrete variants of the Prekopa–Leindler inequality*. Annali della Scuola Normale Superiore di Pisa 22 (2021), 1207–1232. Source-imported discrete midpoint input; its external proof was not freshly audited.

[7] M. Subbarao. Private original manuscript, *TR8-lin-non-lin*, local RamNonlin-IE-PDE-astra corpus. Freshly read lines 65–197, especially coercive inverse, boundary lifting and PDE–Green equivalence. Confidential, not uploaded.

[8] M. Subbarao. Private original Ram Calculus book, volume 21, chapter *Physical metrics*, file `ch01d-physical-metrics.tex`. Freshly read lines 1–210. Confidential, not uploaded.